10 · Pipes

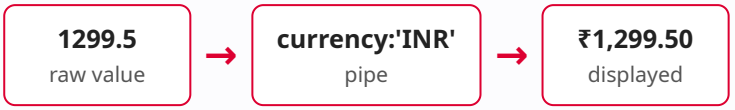
Beginner Angular learning notes

a. Definition

A **pipe** transforms a value for display, without changing the original data. You write it in the template with the `|` symbol: `{{ name | uppercase }}`. Pipes can take arguments after a colon and can be chained.

Angular has many built-in pipes: `date`, `currency`, `percent`, `uppercase`, `lowercase`, `slice`, `json`, `async` and more. You can create a **custom pipe** with the `@Pipe` decorator and a `transform` method.

b. Diagram



A pipe sits between the raw value and the screen

c. Syntax

```
{{ value | pipeName }}
{{ value | pipeName:arg1:arg2 }}
{{ today | date:'dd MMM yyyy' | uppercase }}   // chained

@Pipe({ name: 'myPipe' })
export class MyPipe implements PipeTransform {
  transform(value: string, arg = 1): string { return ...; }
}
```

d. Example Program

`Programs/10_Pipes/pipes-demo.component.ts`

```
import { CurrencyPipe, DatePipe, JsonPipe, PercentPipe, SlicePipe, UpperCasePipe } from
'@angular/common';
import { Component, Pipe, PipeTransform } from '@angular/core';

// A custom pipe: shortens long text
@Pipe({ name: 'truncate' })
export class TruncatePipe implements PipeTransform {
  transform(value: string, limit = 10): string {
    return value.length > limit ? value.slice(0, limit) + '...' : value;
  }
}

@Component({
  selector: 'app-root',
  imports: [CurrencyPipe, DatePipe, JsonPipe, PercentPipe, SlicePipe, UpperCasePipe,
TruncatePipe],
  template: `
    <p>Upper: {{ course | uppercase }}</p>
```

```
    <p>Price: {{ price | currency: 'INR' : 'symbol' : '1.2-2' }}</p>
    <p>Date: {{ launch | date: 'dd MMM yyyy' }}</p>
    <p>Discount: {{ discount | percent }}</p>
    <p>First two: {{ cities | slice: 0 : 2 }}</p>
    <p>Truncated: {{ about | truncate: 14 }}</p>
    <p>Chained: {{ launch | date: 'MMMM' | uppercase }}</p>
    <pre>{{ student | json }}</pre>
  `,
})
export class PipesDemoComponent {
  course = 'angular pipes';
  price = 1299.5;
  launch = new Date(2026, 0, 26); // 26 January 2026
  discount = 0.256;
  cities = ['Chennai', 'Madurai', 'Trichy', 'Salem'];
  about = 'Pipes transform values for display';
  student = { name: 'Arun', marks: [82, 91] };
}
```

e. Output

Rendered in the browser (text on screen):

```
Upper: ANGULAR PIPES
Price: ₹1,299.50
Date: 26 Jan 2026
Discount: 26%
First two: Chennai,Madurai
Truncated: Pipes transfor...
Chained: JANUARY
{
  "name": "Arun",
  "marks": [
    82,
    91
  ]
}
```

f. Real-World Example (English + தமிழ்)

English

A vegetable shop keeps tomato price as 40 in the ledger but prints '₹40.00 per kg' on the bill. The data is unchanged; only the display format changes. That is a pipe.

தமிழ்

காய்கறிக் கடையில் தக்காளி கிலோ ₹40 என்று கடைக்காரர் மனதில் இருக்கிறது. ஆனால் பில்லில் 'தக்காளி - 2 கிலோ - ₹80.00' என்று அழகாக வடிவமைத்து எழுதுகிறார். தக்காளியின் உண்மையான விலை மாறவில்லை; காட்டும் விதம் மட்டுமே மாறியது.

Pipe அதே வேலையைச் செய்கிறது. தரவு (data) அப்படியே இருக்கும்; திரையில் காட்டும்போது தேதி, பணம், சதவீதம் என்று அழகாக மாற்றிக் காட்டும்.

g. Pros

- Reusable formatting
- Pure pipes are cached
- Keeps components clean

h. Cons

- Impure pipes can slow the app
- Locale data must be configured for other regions

i. Advanced / New Features

- All built-in pipes are standalone
- `toSignal` often replaces the `async` pipe
- `@let` can store pipe results

j. Common Mistakes

- Forgetting to import the pipe (e.g. `CurrencyPipe`) in a standalone component's `imports`.
- Using a pipe to change the original data; pipes only change what is displayed.
- Making a pipe impure or heavy; calling slow code inside `transform` runs often.
- Wrong arguments order for `currency` (`code:display:digitsInfo`).

k. Interview Questions

Q1. What is a pipe?
A template feature that transforms a value for display, using `|`.

Q2. Pure vs impure pipes?
Pure pipes run only when their input reference changes (default, fast); impure pipes run on every change detection.

Q3. How do you create a custom pipe?
Use `@Pipe({name})` on a class implementing `PipeTransform` with a `transform` method, and import it where needed.

Q4. Pipe vs a method call in the template?
Pure pipes are cached and only recalculated when inputs change; a method call runs on every change detection.

l. Practice Exercises

1. Display a list of prices in INR using `currency` and sort them in the component.
2. Write a custom `initials` pipe: 'Arun Kumar' -> 'AK'.
3. Show today's date in three different formats using the `date` pipe.